SPL-2 Project Proposal Form, 2026
Institute of Information Technology (IIT)
University of Dhaka

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Project Title:
HistoPath AI: An Integrated Clinical Histopathology Analysis Platform
Project Description:
This project aims to develop a web based platform for AI assisted analysis of H&E stained histopathology images of urinary bladder urothelial carcinoma. It will support case and slide management, image quality assessment, tissue segmentation, feature extraction, cancer classification, grading and mutation detection, explainable AI, pathology reporting and collaboration among laboratory technicians, pathologists, and oncologists. AI outputs will remain preliminary and require review by qualified professionals.
Key Features:
- Role based user account and access management
- Pathology case registration and histopathology slide upload
- Image quality assessment and validation
- Background subtraction and tissue segmentation
- Image tile generation and feature extraction
- AI based cancer classification and tumor grading
- Explainable AI
- Pathologist review, image annotation and result verification
- Cancer staging information management
- Molecular mutation information management
- Pathology report generation, review and approval
- Oncologist case review and clinical collaboration
- LLM based documentation and knowledge assistance
- Workflow tracking and user notifications
- System administration, access control and audit management

Languages or Tools to be used:
Proposed stack: Python, PyTorch, CUDA, OpenCV, JavaScript/TypeScript, React, HTML5/CSS3, FastAPI, PostgreSQL and Git. AI methods: Otsu thresholding, U-Net, ViT, attention/MIL and Grad-CAM. LLM-based documentation assistance may be considered as an optional component.

Supervisor's Name:
Signature of the supervisor:_____
Date:_____

Proposal Presentation Feedback:

Before the Midterm Presentation:
I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.
Signature of the supervisor:_____
Date:_____

Midterm Presentation Feedback:

Before the Final Presentation:
I confirm that the progress is satisfactory and I am forwarding it for the final presentation.
Signature of the supervisor:_____
Date:_____